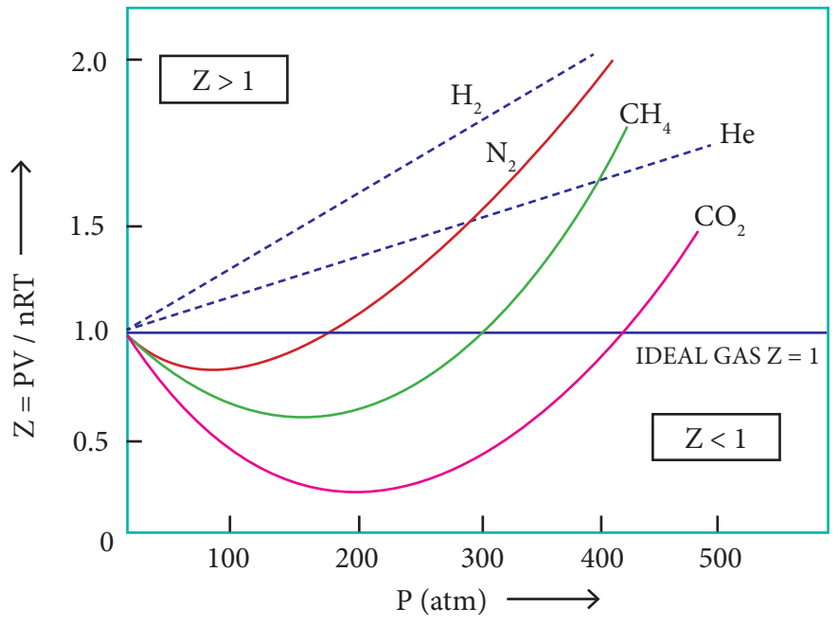

Figure. 6.8 Plot of compressibility factor(Z) vs pressure for some common gases

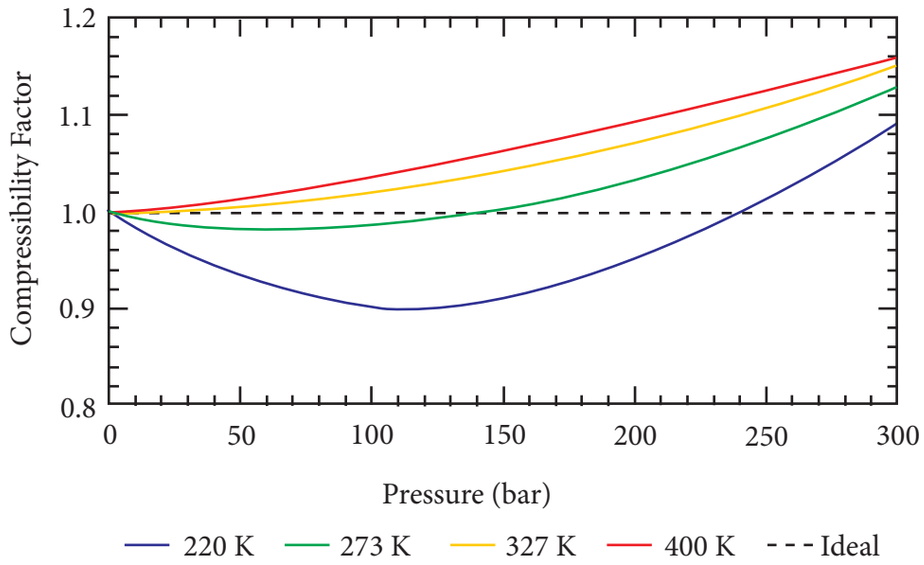

Figure. 6.9 Compressibility factor vs pressure at different temperatures for Nitrogen

When the pressure is low, the volume of the container is very large compared to the volume of the gas molecules so that individual volume of the gas molecules can be neglected. In addition, the molecule in a gas are far apart and attractive forces are negligible. As the pressure increases, the density of gas also increases and the molecules are much closer to one another. Hence, the intermolecular force becomes significant enough to affect the motion of the molecules and the gas will not behave ideally.

At high temperatures the average kinetic energy of the molecules is very high and hence inter moleclular attractions will become insignificant. As the temperature

11th Std Chemistry 158-185.indd 170 06-01-2021 15:35:20